



Islamic University Of Technology

EEE 4308 : Digital Electronics

Project Name : SAP – 1

Team : Digital Electronauts

Phase - D

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Team Members:

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Objective:

- 1) Learning about logical unit of SAP-1
- 2) Learning about Arithmetic unit of SAP-1
- 3) Gaining knowledge about Binary Adder
- 4) Integration with BUS
- 5) Making Output Unit

Description:

This project is about understanding of 6 bit computer and construction of it. Basically its about understanding inside process of a computer. We can understand what does it mean clock speed of a computer and how we can control this speed. For running this computer we need to program the RAM first, then we can run the computer. Program is stored inside RAM as a set of instruction. Other modules acts according to instruction. Program counter counts the instruction and MAR sends the memory address for next instruction. Instructed operation is occurred inside ALU. Value of ALU is stored inside Accumulator. When necessary , Accumulator sends the data to output register to show the result. Necessary signal for all the module is sent by controller sequencer. Controller sequencer generates signals according to instruction code sent by instruction register. All the module is runed by clock pulse. Without clock pulse this computer can't work. There is HLT operation in our computer that stopes the main clock, as a result whole computer stopes. This is total working system of our SAP-1.

Phase - D

1) ALU:

It is the logical unit of our SAP. It performs the logical arithmetic operation. It performs two operation: ADD, SUBTRACT.

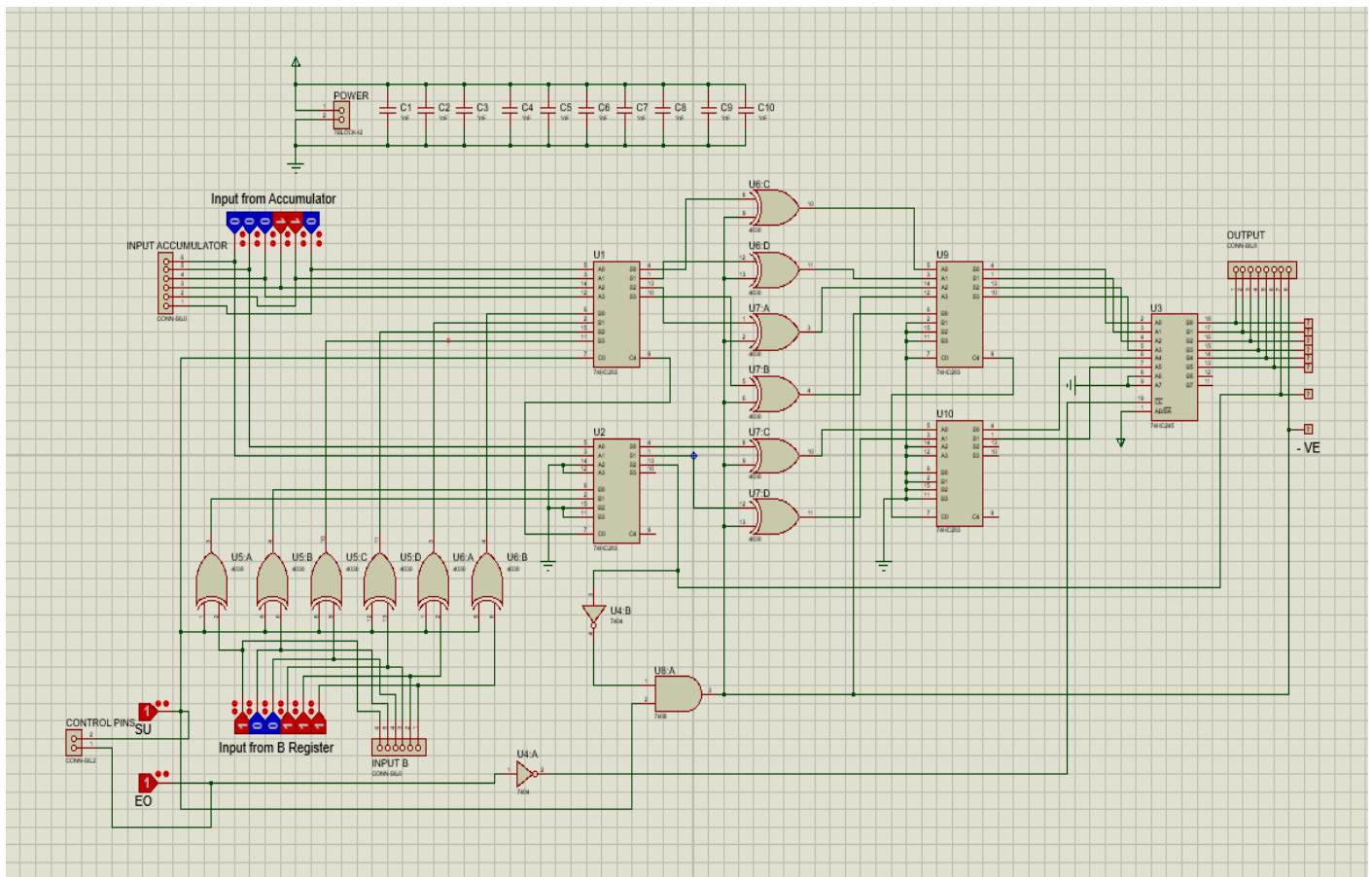
Control Bits:

- 1) **SU**: When it is 0, performs ADD and performs SUB when 1
- 2) **EO**: Sends value to BUS

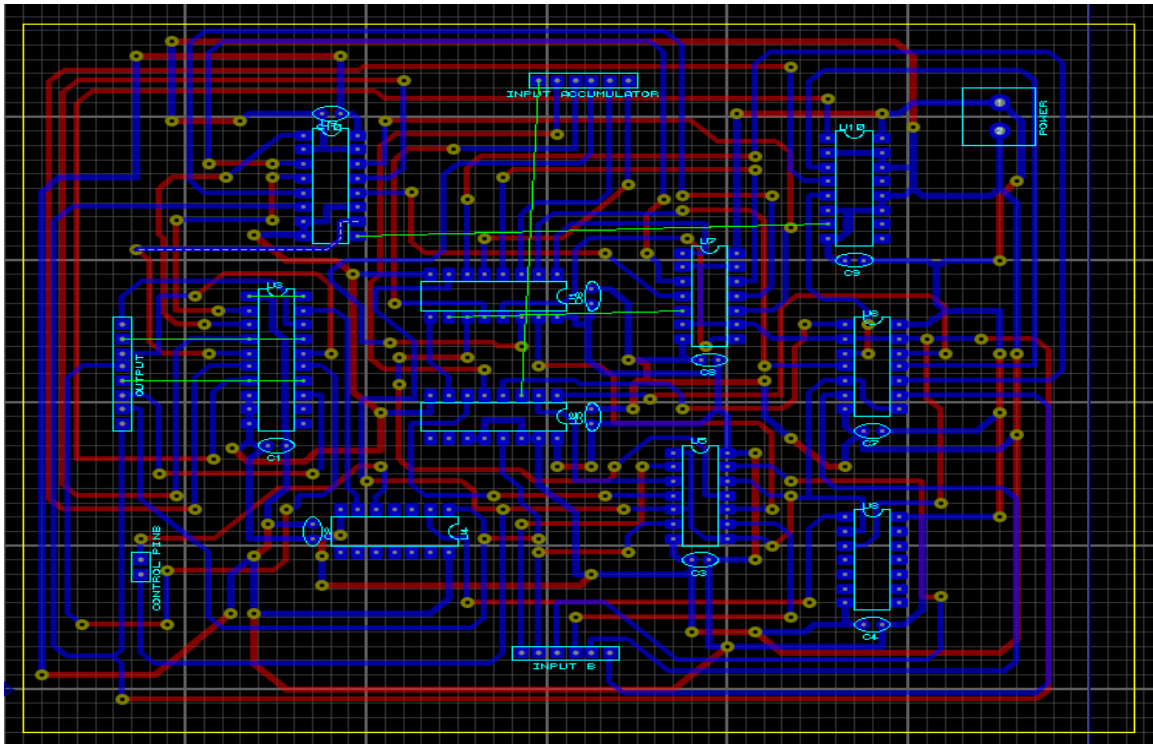
Circuit:

To built this circuit we are using two 74LS173 IC's , one 74LS245 and two 7404 (NOT Gate). 74173 is a 4-bit register and 74245 is Octal BUS transceiver (Tri-state Buffer).

ALU Circuit in Proteous:



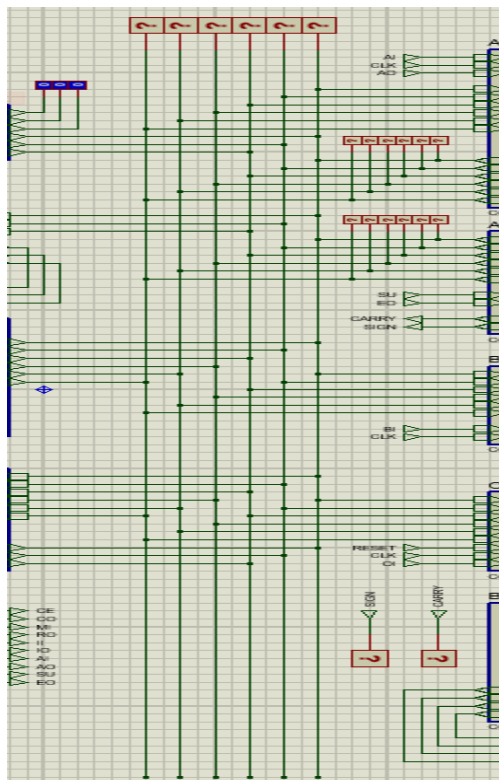
PCB :



2) BUS:

It is six parallel line wires. It is used to combined all module all together.

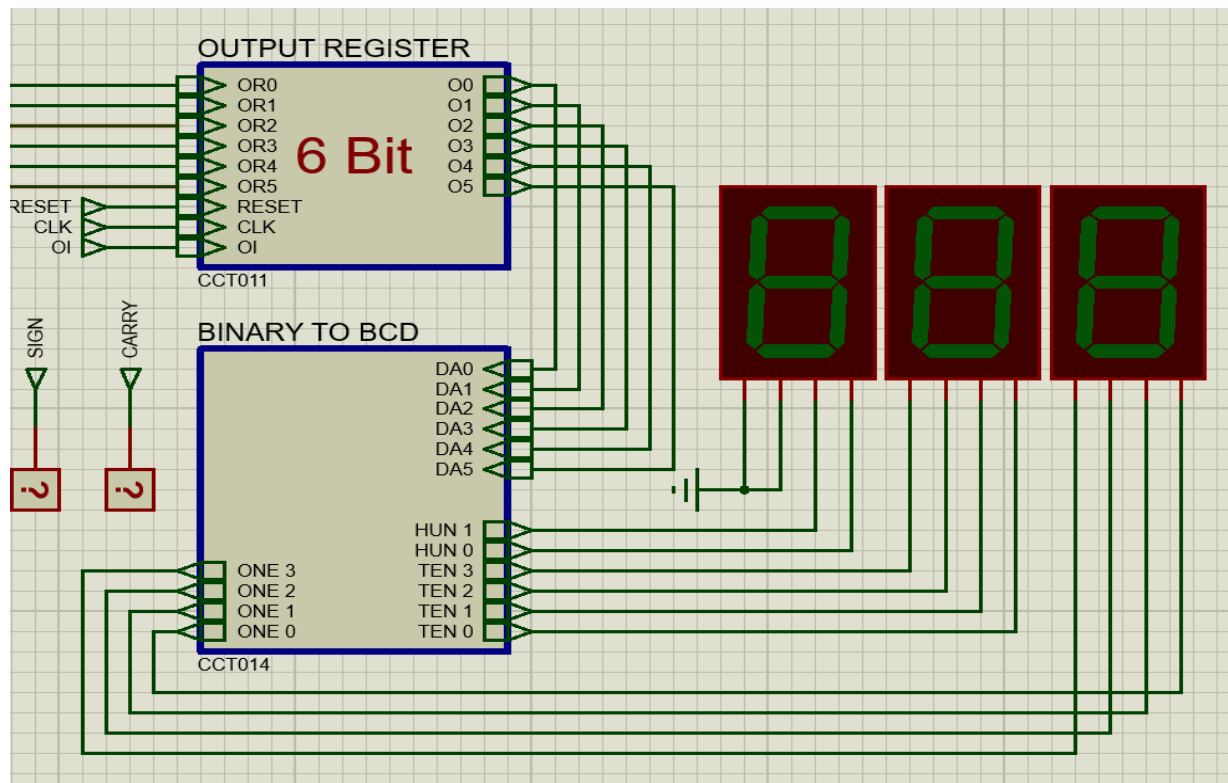
Bus in proteous:



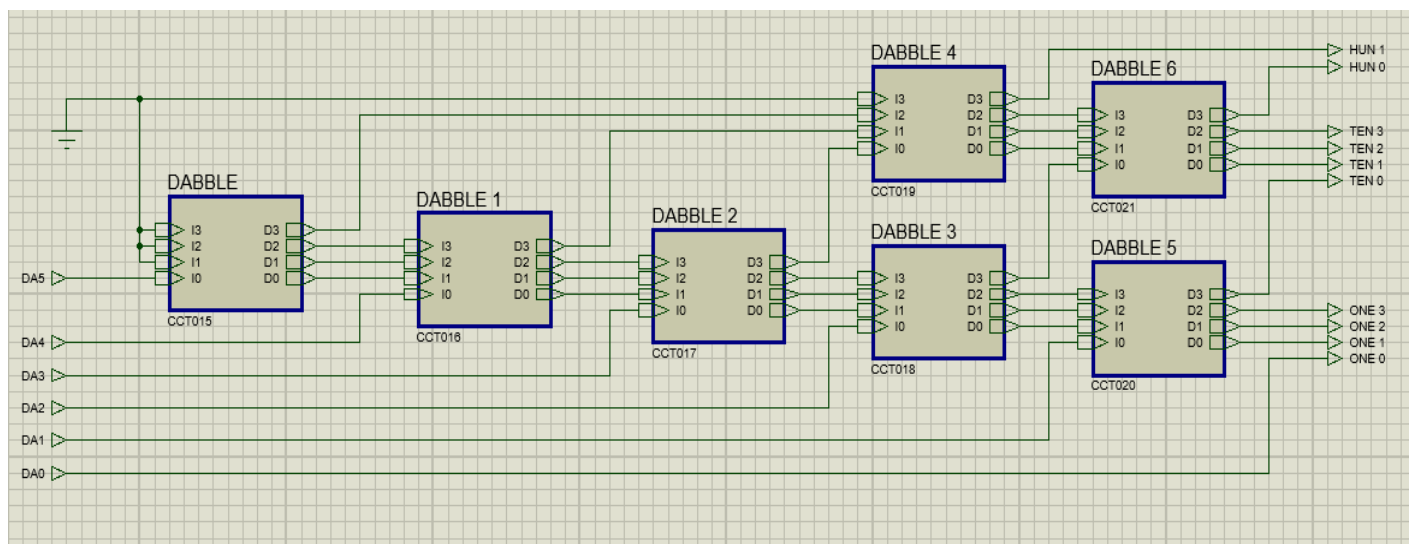
3) OUTPUT Unit:

It is used to show the operation result in a display.

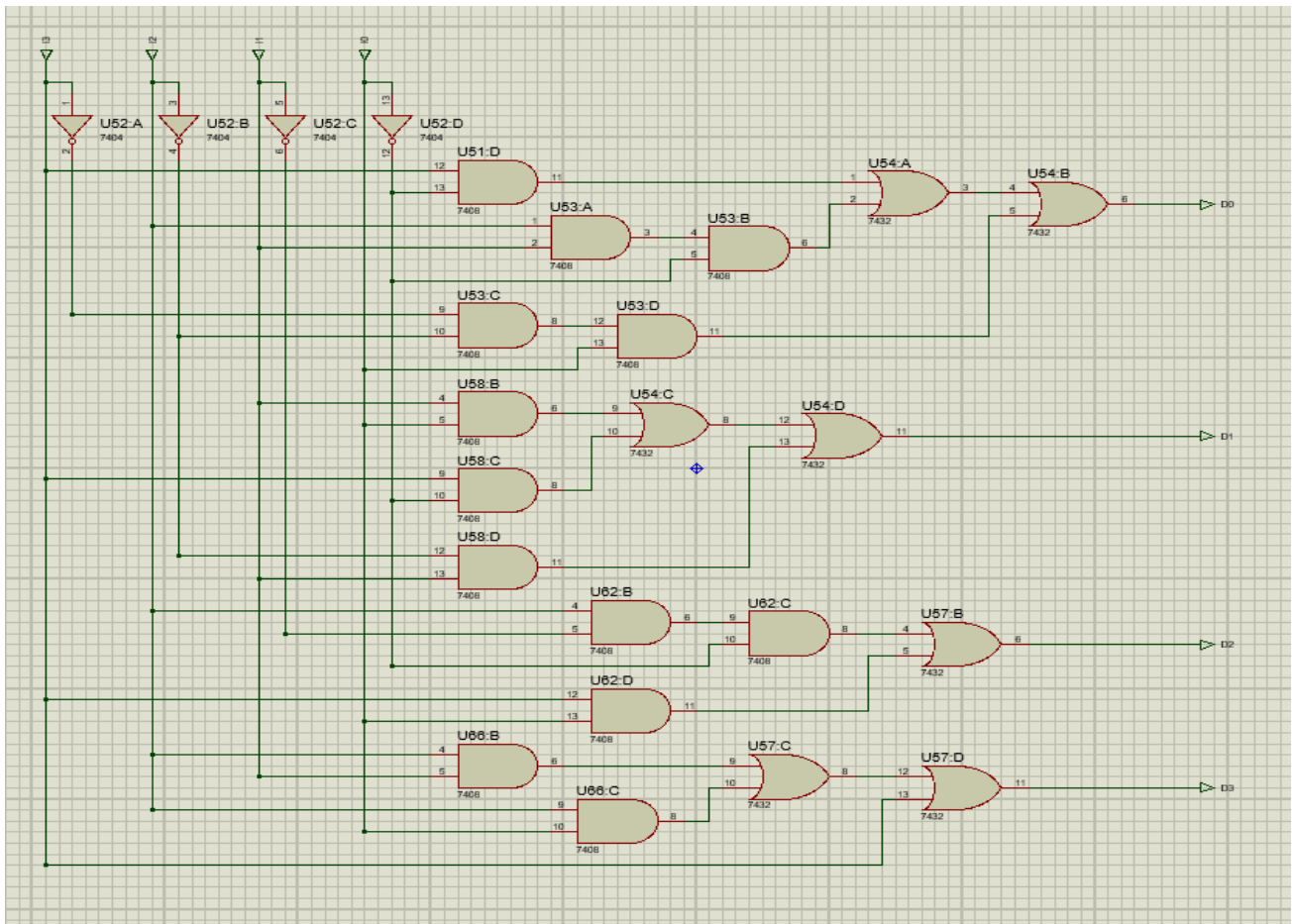
Circuit in Proteous:



Child sheet of Binary to BCD converter:



Dabble circuit:



Problem faced:

- 1) Selecting the proper IC's
- 2) Component placement
- 3) PCB design
- 4) Some components don't have PCB package
- 5) While routing the PCB

Discussion:

In this phase we designed ALU, Output unit and integrate them with 6 bit bus. Among all the module ALU was the tricky one. It is a 6-bit SAP, we can develop it further to 8 bit, as we same number of components and concepts to built 8 bit SAP. Through this phase we learnt how controller sequencer generates necessary signals for the all other module. It was a great learning phase for us.

Contribution of each member is attached with phase-E report.